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SUMMARY

Senior Data Scientist, **AI Architect**, Applied AI Leader, Founder, and Researcher with **14+ years** across Microsoft and SAP designing large-scale data systems, enterprise analytics platforms, experimentation frameworks, and *production-oriented AI workflows*. **7+ years** in Windows data science supporting product experiences reaching **800M+ devices**, with ownership of executive-facing metrics, forecasts, roadmap inputs, AI feature measurement, and adoption strategy. Hands-on with **Python, SQL, PySpark, Azure Databricks**, Azure Synapse, Azure ML, Power BI, Docker, Git-based workflows, **Microsoft Copilot tooling**, LLM-assisted development, **custom MCP servers, multi-agent systems**, and **evaluation frameworks**. Known for translating ambiguous business problems into practical AI/data architecture, trusted metrics, governance patterns, and *adoption-ready systems*.

CORE TECHNICAL SKILLS

AI Architecture, LLM Systems & Microsoft AI Tooling: LLM-assisted workflows, applied generative AI, custom agents, multi-agent teams, MCP servers, custom MCP server development, skills/hooks, prompt orchestration, retrieval workflows, Microsoft Copilot CLI, Copilot Chat, internal Microsoft agency plugins, Athena NLP query workflows, Claude Opus, CLINE, custom evals, reliability gating, AI workflow design.

Databricks, ML & MLOps-Relevant Experience: Azure Databricks, PySpark/Spark-oriented pipelines, Python ML workflows, feature engineering, model evaluation, classification, forecasting, anomaly detection, collaborative filtering, segmentation, churn analysis, model monitoring concepts, lineage, governance, and production-readiness patterns.

Data Engineering, BI & Enterprise Platforms: Advanced SQL, Python, ScopeScript, R, Azure Synapse, Azure Data Studio, Azure ML Studio, Power BI, Jupyter, ETL/ELT pipelines, data modeling, analytics engineering, API integration patterns, SAP HANA, SAP BusinessObjects BI Platform, SAP Data Services, CRM/ERP data, legacy-system migration, data source audits, Jenkins, Git, Docker.

Product Analytics, Strategy & Executive Decision Support: Microsoft Account MAD, DAD, and MAU; Click-to-Do Weekly Engagement Rate and onboarding/adoption metrics; File Explorer Search Success Rate v1; AI Value Index metrics; experimentation strategy; forecasting; roadmap prioritization; SLT/CVP/VP-level executive communication.

Leadership, Governance & Enablement: AI roadmap definition, responsible AI/evaluation concepts, data governance, metric lineage, documentation, training/onboarding, stakeholder trust-building, vendor/partner collaboration, intern management, cross-functional leadership, privacy/compliance review, product ownership, user research, prototyping, experimentation, and launch readiness.

APPLIED AI, ARCHITECTURE & GOVERNANCE

- Designed and implemented **LLM-assisted workflow pipelines** for structured decision support, including **prompt orchestration, evaluation scoring, reliability gates**, reusable review loops, and decision-state tracking.
- Built **agent-style architectures** integrating structured tools, retrieval workflows, **custom MCP servers**, skills, hooks, and **multi-agent collaboration patterns**.
- Developed **custom evaluation harnesses** from scratch to assess **output correctness**, consistency, quality thresholds, and **workflow reliability** across AI-assisted systems.
- Applied **CLINE, Claude Opus, Copilot CLI, Copilot Chat**, internal Microsoft agency plugins, **Athena NLP query workflows**, and AI-assisted development practices to accelerate prototyping, data analysis, documentation, and repository creation.
- Evaluated when to use **traditional ML**, metrics/automation, **generative AI**, or **human-in-the-loop workflows** depending on reliability needs, business impact, data quality, and *stakeholder adoption risk*.

PROFESSIONAL EXPERIENCE

Founder & Technical Lead - HitsHome

AI Decision Intelligence Platform - Real Estate & Location Analytics | 2024-Present

- Architecting an **AI-powered decision intelligence platform** integrating **structured data pipelines**, scoring models, **LLM-assisted workflows**, and agent-enabled decision support for real estate and location analytics.
- Designed **explainable scoring systems** to evaluate geographic investment potential using **multi-source economic, demographic, housing, and market datasets**, aligning technical architecture to investor, homebuyer, and market decision needs.
- Built scalable **data ingestion and feature engineering pipelines** combining Census, housing, and economic data into **standardized decision-ready features** across counties, places/cities, ZCTAs/ZIP codes, and state-level geographies.
- Developing **personalized market analysis**, property matching, **structured risk assessment**, confidence scoring, and **explainable recommendation workflows** using agentic systems, schemas, custom tools, and evaluation patterns.
- Created **product strategy**, data model concepts, **roadmap artifacts**, source sufficiency rules, stale-data windows, and **governance concepts** for an AI-enabled analytics platform designed to make complex location decisions clearer and more actionable.
- Used **AI-assisted development workflows, custom agents**, and **evaluation patterns** to bootstrap repositories, documentation, technical plans, and implementation roadmaps for a real estate intelligence platform.

Senior Data Scientist - Microsoft

Windows Operating System, Modernize AI Team | 2022-Present

- Drive YoY growth for the **Microsoft Account Key Result** portfolio, including **MAD, DAD, and MAU** measurement; deliver recurring leadership updates, **annual forecasts**, target-setting analysis, anomaly detection, root-cause analysis, and **roadmap recommendations**.
- Owned forecast inputs and informed/prioritized annual roadmaps from **FY21 through FY26** across Microsoft Accounts and later **Click-to-Do**, translating metric movement, adoption patterns, and product hypotheses into **investment decisions**.
- Presented Microsoft Accounts data in **Windows Business Review** forums for **5+ years** to org-wide **SLT audiences including CVP- and VP-level leaders**, balancing technical accuracy, business impact, and clear executive communication.
- Spearhead **metric definition alignment** and logic harmonization with partner organizations, improving consistency, **lineage**, and **decision quality** across Windows measurement surfaces.
- Designed and built **File Explorer Search Success Rate v1**, a **custom multi-component metric framework** measuring search success, click-through quality, click-back behavior, result position weighting, and task-completion quality.
- Supported analytics, onboarding/adoption measurement, and **Weekly Engagement Rate** evaluation for **Click-to-Do**, an **AI-powered Windows feature**; identified priority funnel drop-offs and adoption opportunities.
- Created **AI Value Index metrics** and related frameworks to evaluate AI experiences across product usage, engagement, success signals, and business value, supporting **roadmap prioritization** for **emerging AI features**.
- Applied **collaborative filtering, time-series forecasting, anomaly detection, user segmentation**, classification, churn analysis, and user-flow onboarding analysis to identify growth opportunities and improve product decisions.
- Redesigned Windows account measurement from **device-based to user-based attribution** by **creating and validating the data pipeline**, improving metric accuracy, sensitivity, and decision usefulness across partner organizations.
- Envisioned and advanced a new **account-profile gamification feature space**, including avatar badging, validating feature-impact hypotheses through experimentation and influencing **10+ initiatives** adopted across Windows 11 surfaces; tied to a projected **600M+ revenue/business opportunity**.
- Initiated and coordinated additional functional IC resources for the **FY24 roadmap** focused on **personalization** and deeper use of the user-based data product, using data insights and feature gains to justify **expanded investment**.

Data and Applied Scientist II, Windows OS - Microsoft

Windows Operating System, Core Windows Experiences | December 2018-January 2022

- Functional manager and project lead for a **team of 3** through the **full data science lifecycle**: hypotheses, user research, prototyping, testing, implementation, experimentation, and retail feature launch.
- Provided **coaching and feedback**, completed performance review cycles, coordinated with university programs, and was trusted to **recommend full-time employment offers** to all three interns.
- Managed data projects across organizations to increase engagement and revenue through the **Microsoft Account Attach Rate** metric and sub-metrics, **machine learning clustering**, exploratory insights, and **multi-source data joins**.
- Vetted and led **experimentation shipment processes**, finalized green-light launch recommendations with leadership, and used statistical analysis for **scorecard creation** and high-quality **hypothesis testing**.
- Built analysis and reporting workflows using **Python, ScopeScript, Azure ML, Azure Data Studio, SQL, Power BI**, and supervised/unsupervised data science methods.

Research Assistant - Microsoft Research

Ability Lab, Neurodivergence and Multimodal AI Design | January 2019-October 2020

- Ideated and generated **hypotheses**, crafted a research vision, conducted **experiments and interviews**, **coded and analyzed results**, and wrote/edited academic text for Microsoft Research Ability Lab collaborations.
- Focused on **cognitive information processing** and **communication models in autistic populations**, contributing to a new framework and model with Microsoft Research collaborators.
- Contributed to **published HCI and accessibility research** presented in ACM venues, connecting qualitative insight, experimental design, **responsible AI** questions, and **human-centered product design**.

Data Scientist; QA Developer; Product Owner - SAP

Enterprise Analytics | April 2015-October 2018

- Applied **CRM and ERP system expertise** to **predictive modeling** using SAP Predictive Analytics, Python, R, and Anaconda; built data models, **ETL workflows**, and reports in **SAP HANA**.
- Architected **migration strategy** for SAP BusinessObjects Platform, including audit of **200 data sources and 40K reports**; managed messy **legacy-system dependencies** and reduced enterprise reporting migration risk.
- Implemented BusinessObjects Cloud Library Extension **automated versioning updates** in the main product code line and created a **regression-identification framework** using **Jenkins and Docker**.
- Implemented **Agile model** for Build Infrastructure team and decommissioned **77 servers**, representing approximately **80% of internal DevOps application servers**, including backup strategy, execution, and communication.

- Led creation of a **full-stack analytic solution** for **SAP Product Idea Management**, extracting customer feature request data from BrightIdea via PHP, Python, Selenium, and SAP Data Services into SAP HANA for analytics serving approximately **240 SAP product managers** and **50+ SAP products**.
- Served as **intern manager** and **hiring/onboarding lead** for new full-time team members.

Global Project Lead; BI Tools Engineer; Database Developer - SAP

Development Operations | September 2011-February 2015

- Led **system upgrades**, testing cycles, **performance enhancements**, and customer support for SAP BusinessObjects Enterprise Platform 4.1 on a globally used internal instance serving **2000+ users**.
- Proactively addressed system warnings to reduce customer impact, using **Zabbix** and **Windows PerfMon** for monitoring and diagnosis.
- Led **post-upgrade automated testing** through data extraction, modeling, and reporting workflows using **Java and Selenium**.
- Led **three co-op hiring rounds** with four competitive hires and led an SBOP global project **team of six** on an ongoing basis.
- Supported DevOps unit through **troubleshooting**, administration, data extraction, **security**, monitoring, and **data modeling** across SBOP, IBM ClearQuest, **JIRA**, and Perforce CVS.
- Improved **event monitoring** through **audit database creation** using SAP BusinessObjects IDT.

LEADERSHIP, COMMUNITY & CULTURE FIT

- **6+ years** as **team lead**, **intern manager**, mentor, and project lead; conducted interviews, managed hiring/onboarding, supported performance reviews, and led **teams of 2-6** contributors across Microsoft and SAP contexts.
- **8+ years** of product owner and analyst experience, including **roadmap definition**, stakeholder requirements, **metric alignment**, compliance considerations, risk communication, and data-driven product execution.
- **Co-Founder, CEO, and President** - Girls in Tech Vancouver: led strategic development and operations, managed **20+ volunteers**, served as board president, grew member base to **400+**, secured **30+ sponsors**, and orchestrated **20+ events** to support retention of women in technology.
- **Founder** - SAP-UBC GIRLsmarts4tech: established SAP and UBC partnership teaching computer science to Grade 7 girls; contributed **400+ leadership volunteer hours**; grew volunteer base from **5 to 150** in 1 year; supported **seven program runs**, including three international iterations.
- **3000+ hours** extracurricular volunteering across Paul Allen Institute consciousness research, Windows Inclusivity Culture Team, UBC-SAP GIRLsmarts4tech, Microsoft Research Ability Team, Bringing Women Together Network, UBC Sauder mentoring initiatives, **design-thinking coaching**, youth education, and **community programs**.

PUBLICATIONS & RESEARCH

- **"Managing Stress: The Needs of Autistic Adults in Video Calling"** - Proceedings of the ACM on Human-Computer Interaction, Volume 3, Issue CSCW, November 2019.
- **"Lessons Learned in Designing AI for Autistic Adults"** - ASSETS 2020: Proceedings of the 22nd International ACM SIGACCESS Conference on Computers and Accessibility, October 2020.
- Research in coherence in **rodent MRI data** - Paul Allen Institute consciousness and psilocybin research collaboration.
- **Computational Neuroscience** Research Topics graduate course - Computational Neuroscience Center, University of Washington; **top of class**.

EDUCATION

- 2022 - **University of Washington**, United States - Computational Neuroscience Course.
- 2018 - **BrainStation**, Canada - Data Science and Modeling.
- 2010-2013 - **University of British Columbia**, Canada - **B.Sc., Major in Computer Science** with Co-op. Course highlights: Artificial Intelligence, Calculus, Linear Algebra, Databases, Cognitive Science, and Statistics.
- 2008-2010 - **University of Victoria**, Canada - B.Sc. studies in Computers, Mathematics, and Economics; transferred.